

When 3% means nothing: calibrating escalation limits to a bank's own forecasting error distribution.

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Motivation

ROUND NUMBERS AS RISK GOVERNANCE

- Bank-internal governance frameworks escalate when a forecasting error, of a balance-sheet item, of net interest income, of a stress-test projection, exceeds a threshold. In practice the threshold is a round number, commonly three percent, imposed uniformly across institutions and portfolios.
- A fixed threshold has no distributional content: whether a 3% error is routine noise or a genuine signal depends entirely on the institution's own forecasting error distribution.
- The paper proposes and evaluates calibrating escalation limits to that distribution.

The calibration rule

THRESHOLDS AS QUANTILES

For institution i with empirical forecasting-error distribution F_i of the monitored quantity, the escalation limit at tolerance α is the quantile

$$c_{i,\alpha} = F_{|e|,i}^{-1}(1 - \alpha), \quad \text{so that} \quad \Pr(|e_{it}| > c_{i,\alpha}) = \alpha.$$

- Under the quantile rule, the escalation frequency is equalised across institutions by construction; under a uniform fixed threshold it is whatever the institution's error dispersion makes it.
- The rule requires only the institution's own forecasting track record, an input every supervised bank already possesses.

Evidence

POLISH BANK STRESS-TEST DATA

- The calibration is developed on Polish bank stress-test forecast data across institutions of different size and macroeconomic exposure.
- A uniform 3% threshold either grossly under-triggers or over-triggers depending on the institution: for a large, diversified balance sheet it fires on routine noise, for a small or macro-exposed institution it misses genuinely informative errors.
- The quantile-calibrated limits restore a common escalation probability and make threshold breaches comparable across the system.

Implications

FOR RISK GOVERNANCE AND SUPERVISION

- Escalation frameworks gain statistical content: a breach carries a known and uniform surprise probability, so it is informative by design.
- Supervisory comparability improves, since breach counts across institutions measure the same tail event.
- The rule composes with the existing governance apparatus; it replaces only the number, an institution-specific quantile in place of a system-wide constant.

Thank you.

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